HanuAi ML Assessment — Submission Report

Prepared using the supplied Task 2 dataset. Findings are exploratory and should be validated before operational use.

Executive Summary

Task 2 includes data-quality checks, exploratory distributions, rule-based text tags, TF-IDF/K-Means clusters, enriched CSV and Excel outputs, and stakeholder recommendations.

Task 1 — Compliance and Sentiment Analysis

Best Buy Canada's current Conditions of Use explicitly prohibit scraping and data-mining. The assignment also requires compliance with website policy, so this package does not automate collection from BestBuy.ca. The Task 1 notebook instead implements cleaning, VADER sentiment scoring, aspect tagging, and CSV export for an authorized review export. No live review data or results are fabricated. Obtain written permission or an approved export to complete the live-data portion.

Source: https://www.bestbuy.ca/en-ca/help/policies-and-terms-and-conditions/conditions-of-use

Task 2 — Analysis and limitations

Use the enriched CSV and Excel workbook for the event-level tags and summaries. Keyword tags can overlap and may miss synonyms or negation. TF-IDF/K-Means clusters are exploratory topic groupings, not confirmed engineering root causes. Compare event counts with exposure denominators before interpreting rates.

Stakeholder recommendations

• Prioritize recurring audio, display, connectivity, and communication themes for engineering review when supported by event counts and case examples.

• Create Pareto views by issue tag, causal code, model year, build plant, and time period; add exposure denominators before comparing risk.

• Separate customer symptoms, technician-confirmed causes, and correction actions in reporting.

• Manually label representative records and evaluate tagging precision/recall before operational deployment.

• Review cluster examples with engineering specialists and monitor recurrence after repairs.